# WiFi Assignment Module - 1

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**Questions**

1. In which OSI layer the Wi-Fi standard/protocol fits.

2. Can you share the Wi-Fi devices that you are using day to day life, share that device's wireless capability/properties after connecting to network. Match your device to corresponding Wi-Fi Generations based on properties

3. What is BSS and ESS?

4. What are the basic functionalities of Wi-Fi Accesspoint

5. Difference between Bridge mode and Repeater mode

6. What are the differences between 802.1 la and 802.11b.

7. Configure your modem/hotspot to operate only in 2.4Ghz and connect your laptop/ Wi-Fi device, and capture the capability/properties in your Wi-Fi device. Repeat the same in 5Ghz and tabulate all the differences you observed during this

8. What is the difference between IEEE and WFA

9. List down the type of Wi-Fi internet connectivity backhaul, share your home/college's wireless internet connectivity backhaul name and its properties

10. List down the Wi-Fi topologies and use cases of each one.

Solutions:

1. Wi-Fi (IEEE 802.11) operates at the **Physical Layer (Layer 1)** and the **Data Link Layer (Layer 2)** of the OSI model. At the Physical layer, it defines how data is transmitted over radio frequencies , modulation techniques, frequency bands, signal encoding, and transmission power. At the Data Link layer, specifically the **MAC (Medium Access Control) sublayer**, it handles how devices share the wireless medium, manages frame formatting, handles access control using CSMA/CA (Carrier Sense Multiple Access with

Collision Avoidance), and deals with association, authentication, and acknowledgment of frames. The upper sublayer of Data Link , LLC (Logical Link Control) , remains common with wired Ethernet, which is why Wi-Fi can seamlessly bridge into a wired network.

2. The WiFi device that I use in my day to day life is my personal laptop, its wireless properties when connected to wifi is shown below.

| | |
|---|---|
| SSID: | Airtel_F1 |
| Protocol: | Wi-Fi 6 (802.11ax) |
| Security type: | WPA2-Personal |
| Manufacturer: | MediaTek, Inc. |
| Description: | MediaTek Wi-Fi 6 MT7921 Wireless LAN Card |
| Driver version: | 25.40.2.579 |
| Network band (channel): | 5 GHz (52) |
| Aggregated link speed (Receive/ Transmit): | 1201/1201 (Mbps) |

The protocol is 802.11ax, belongs to WiFi 6/6E generation, can offer maxspeed of 9.6Gbps, and can operate in 2.4/5/6 Ghz band.

3. **BSS (Basic Service Set)** is the most fundamental building block of a Wi-Fi network. It consists of a single Access Point and all the wireless clients associated with it. Every BSS is identified by a **BSSID**, which is essentially the MAC address of the AP's radio interface. When you connect your phone to a single home router, you are part of one BSS. A BSS can also exist without an AP in what is called an **IBSS (Independent BSS)** , commonly known as an ad-hoc network , where devices communicate directly with each other without any AP.

   **ESS (Extended Service Set)** is formed when two or more BSSs are connected together through a common wired backbone called the **Distribution System (DS)**, typically Ethernet. All APs in an ESS share the same **SSID**, so from the user's perspective it looks like one continuous network. As a user moves across the coverage area, their device can roam from one AP to another without manually reconnecting. A university

campus Wi-Fi or a corporate office Wi-Fi with multiple APs all broadcasting the same SSID is a perfect example of an ESS.

4. An Access Point's primary function is to act as a bridge between the wireless network and the wired network. It converts wireless 802.11 frames from clients into Ethernet frames and forwards them to the wired LAN, and vice versa. The AP continuously broadcasts **Beacon frames** at regular intervals (typically every 102.4 ms) to advertise its presence, SSID, supported data rates, and capabilities so that nearby devices can discover and connect to it.

   The AP manages the **association and authentication** process , when a client wants to connect, the AP handles the handshake, verifies credentials (through open auth, WPA2, WPA3, or 802.1X), and assigns the client a place in its association table. It also controls **medium access** using CSMA/CA to ensure multiple clients share the air fairly without constant collisions. In addition, the AP handles **rate adaptation** , dynamically adjusting the data rate based on the signal strength and quality of each connected client. It also performs **VLAN tagging**, **QoS prioritization**, **power saving support** (through buffering frames for sleeping clients), and in managed deployments, it communicates with the WLC for centralized policy enforcement.

5. In **Bridge mode**, the AP or router disables its own NAT and DHCP server and connects to an existing network purely to extend the wired or wireless infrastructure. It acts as a transparent link between two network segments , for example, wirelessly connecting two buildings so they behave as one LAN. Devices connected through a bridge receive IP addresses from the main router, and the bridge itself is nearly invisible to the network. This mode is used when you want to join two separate network segments without creating a new sub-network.

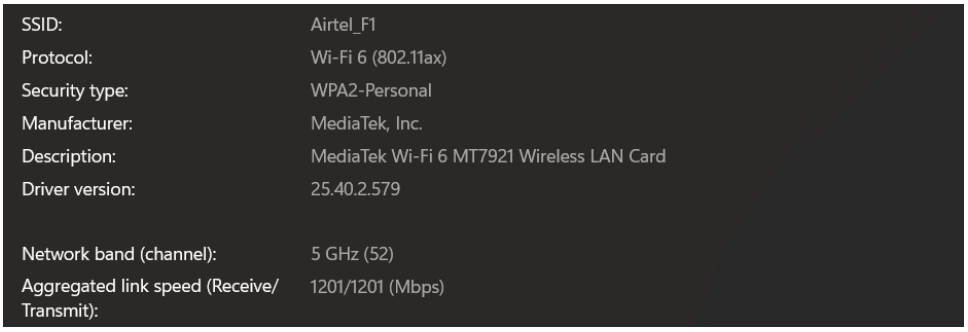
   In **Repeater mode**, the device connects to an existing Wi-Fi network and rebroadcasts the same signal to extend the wireless coverage area. The repeater receives the Wi-Fi signal from the main router and retransmits it, allowing clients that are too far from the main router to connect through the repeater. However, repeater mode has a significant drawback , because the repeater uses the same radio to receive and retransmit, the available bandwidth is roughly halved. It also adds latency. Repeater mode is a simple

and low-cost way to eliminate dead zones at home, but it is not ideal for performance-sensitive environments.

6. 802.11a and 802.11b were both released in **1999** but are fundamentally different standards. 802.11b operates in the **2.4 GHz** band and offers a maximum data rate of **11 Mbps**. The 2.4 GHz band has good range and wall penetration but is heavily congested because it is shared with Bluetooth, microwaves, baby monitors, and neighboring Wi-Fi networks. 802.11b uses **DSSS (Direct Sequence Spread Spectrum)** modulation.

   802.11a operates in the **5 GHz** band and offers a much higher maximum data rate of **54 Mbps**. The 5 GHz band is less congested and provides cleaner, faster communication but has shorter range and weaker penetration through walls and obstacles compared to 2.4 GHz. 802.11a uses **OFDM (Orthogonal Frequency Division Multiplexing)** modulation, which is more spectrally efficient. Because they operate on completely different bands, 802.11a and 802.11b are **not compatible** with each other and cannot communicate directly.

   In 5Ghz



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| Driver version: | 25.40.2.579 |
| Network band (channel): | 5 GHz (52) |
| Aggregated link speed (Receive/ Transmit): | 1201/1201 (Mbps) |

   In 2.4Ghz

| | |
|---|---|
| SSID: | Airtel_F1 |
| Protocol: | 802.11g |
| Security type: | WPA2-Personal |
| Manufacturer: | MediaTek, Inc. |
| Description: | MediaTek Wi-Fi 6 MT7921 Wireless LAN Card |
| Driver version: | 25.40.2.579 |
| Network band (channel): | 2.4 GHz (11) |
| Aggregated link speed (Receive/ Transmit): | 144/144 (Mbps) |

When changed the band from 5Ghz the protocol changed to 802.11g from 802.11ax, the channel changed from 52 to 11 and link speed reduced to 144 Mbps from 1201 Mbps on 5Ghz

7. **IEEE (Institute of Electrical and Electronics Engineers)** is a global technical standards body made up of engineers and researchers. In the context of Wi-Fi, IEEE is responsible for defining and publishing the **802.11 standards** , the technical specifications that describe exactly how wireless communication works at the physical and MAC layers. IEEE defines things like modulation schemes, frame formats, frequency usage, and protocols. IEEE does not manufacture products or certify devices , it purely defines the technical standard through its working groups.

   **WFA (Wi-Fi Alliance)** is an industry consortium of companies including Apple, Intel, Qualcomm, Cisco, and hundreds of others. The Wi-Fi Alliance takes the IEEE 802.11 standards and creates **interoperability certification programs** to ensure that devices from different manufacturers actually work correctly together. When a device passes Wi-Fi Alliance testing, it earns the **"Wi-Fi Certified"** logo. The WFA also brands the generations with friendly names , Wi-Fi 4, Wi-Fi 5, Wi-Fi 6 , to make it easier for consumers to understand. So in short, IEEE writes the rules and WFA ensures manufacturers follow and implement those rules consistently.

8. The backhaul refers to the connection that links the Access Point or router back to the internet or the core network. The main types are listed below.

**Fiber Optic (FTTH/FTTP)**: Fiber to the Home is the fastest and most reliable backhaul. It uses light through glass fiber cables and supports speeds from 100 Mbps to multiple Gbps with very low latency. Most modern urban deployments use this.

**DSL (Digital Subscriber Line)**: Uses existing telephone copper lines to carry internet. ADSL is the most common variant, offering asymmetric speeds (faster download than upload). Speeds typically range from 1 Mbps to 100 Mbps depending on distance from the exchange.

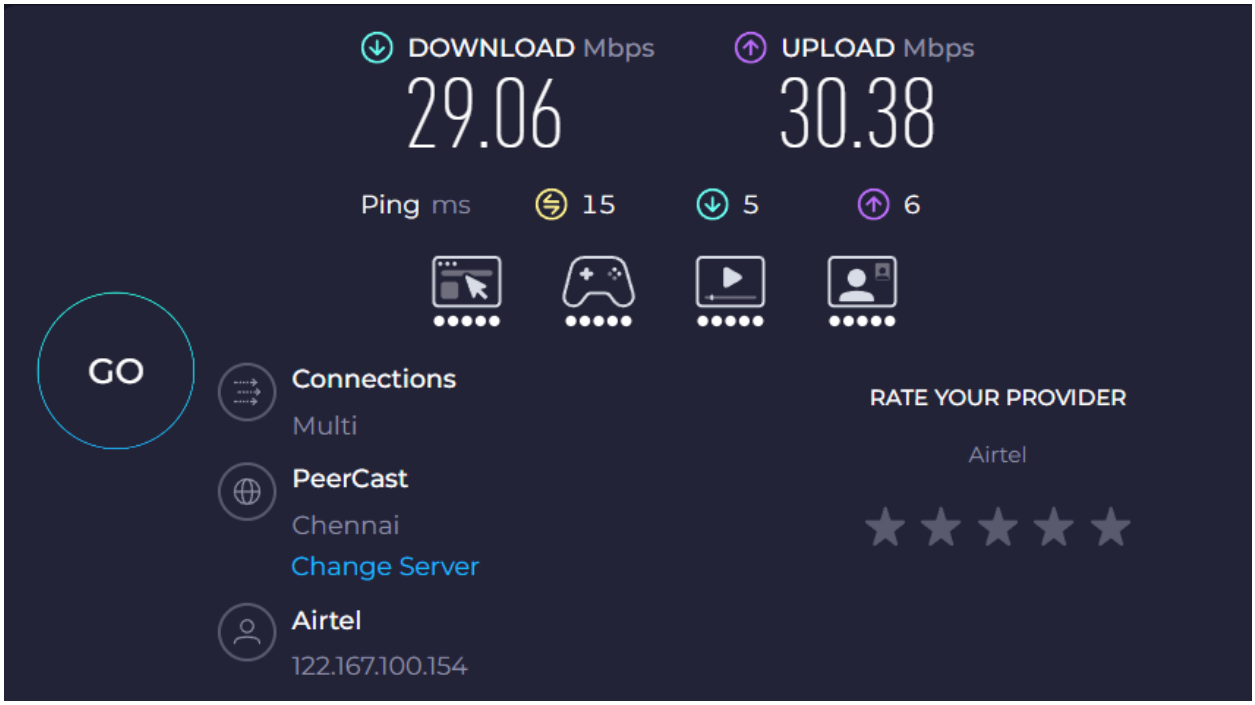
**Cable (DOCSIS)**: Uses coaxial cable TV infrastructure to carry internet. Generally faster than DSL, offering speeds from 50 Mbps to over 1 Gbps. Bandwidth is shared among users in the same neighborhood segment.

**4G/5G Cellular (Fixed Wireless Access)**: A cellular modem or router receives mobile data signals and provides internet to local Wi-Fi devices. Used heavily in rural areas or as a backup connection. 5G FWA can deliver speeds comparable to fiber in some areas.

**Satellite**: Internet delivered via satellite, such as traditional VSAT or modern LEO (Low Earth Orbit) services like Starlink. VSAT has high latency (600ms+), while Starlink brings latency down to 20–50ms. Used where no other backhaul is available.

**Leased Line / MPLS**: Dedicated enterprise-grade connections with guaranteed bandwidth and SLAs. Used in corporate offices, universities, and data centers.

The backhaul I use is FTTH connection by airtel, the fiber optic cable from Airtel's nearest OLT (Optical Line Terminal) on the street connects into my router, which converts the optical signal into electrical signal and then provides you internet over Wi-Fi and LAN ports. This is why your speeds are high enough to support Wi-Fi 6 at 1201 Mbps link rate.




9. **Infrastructure Mode (BSS)** is the most common topology where all wireless clients connect to a central Access Point, which then connects to the wired network. Every client communicates through the AP, not directly with each other. This is used in homes, offices, airports, and virtually all standard Wi-Fi deployments.

   **Extended Infrastructure (ESS)** is multiple APs connected via a wired backbone all sharing the same SSID, allowing seamless roaming across a large area. This is used in universities, hospitals, shopping malls, and corporate campuses where a single AP cannot cover the entire area.

   **Ad-Hoc Mode (IBSS)** is a peer-to-peer topology where devices communicate directly with each other without any AP. There is no central coordination. This is used for quick temporary file transfers between two laptops, or in military/disaster recovery scenarios where no infrastructure exists.

   **Wireless Distribution System (WDS)** allows APs to connect to each other wirelessly to form a network without a wired backbone between them. One AP connects to the

internet and shares it wirelessly to other APs. Used in environments where running Ethernet cable between APs is not feasible.

**Mesh Topology** is an evolution of WDS where multiple APs (mesh nodes) intelligently communicate with each other to form a self-healing, self-organizing network. Traffic finds the best path through the mesh automatically. Used in large homes, outdoor public Wi-Fi, smart city deployments, and warehouses where full wired coverage is impractical.

**Point-to-Point (PtP)** uses two directional antennas to create a wireless link between two fixed locations , for example, connecting two buildings across a campus wirelessly. Used as a backhaul link replacement for situations where laying fiber is expensive or impossible.

**Point-to-Multipoint (PtMP)** has one central AP with a sector or omnidirectional antenna serving multiple remote locations simultaneously. Used by Wireless ISPs (WISPs) to serve multiple subscriber locations from a single tower.